

Fine-Tuning Flan-T5-small with LoRA on DialogSum

DATA 266 — Findings Report | Ritika Mukesh Neema (SID 019306638) | SEED=6638

1. Setup

I fine-tuned **google/flan-t5-small** (about 77M parameters) with LoRA via Hugging Face PEFT on the **DialogSum** dataset (*neil-code/dialogsum-test*: 1,999 train / 499 validation / 499 test). Each example was converted to input = dialogue (with the prefix “Summarize the following conversation.”) and target = human summary. Training used a seeded subset of **638** dialogues (SLICE = 6638 mod 1000); evaluation used the first 100 test dialogues. Runs were on a Google Colab Tesla T4. LoRA was applied to the attention **q** and **v** projections with dropout 0.05 and alpha = 2r; training was 5 epochs, batch size 8, learning rate 1e-3.

2. Results

Model	r / alpha	Trainable params	Final loss	ROUGE-1	ROUGE-2	ROUGE-L
Baseline (no fine-tuning)	-	0	-	0.148	0.047	0.136
LoRA r=4	4 / 8	172,032 (0.22%)	1.534	0.344	0.106	0.282
LoRA r=16	16 / 32	688,128 (0.89%)	1.450	0.350	0.106	0.283

Total parameters: 77,649,280 (r=16) and 77,133,184 (r=4). Each run trained in about 2 minutes on the T4.

3. Before vs. after fine-tuning (same two test dialogues)

	Dialogue 1 (test idx 182)	Dialogue 2 (test idx 442)
Reference	#Person1# and Jack are talking about Jack's favorite classes.	#Person1# and #Person2# are blaming each other for lots of bad experience during the journey.
Baseline	W: Hi, Jack. How were your classes this semester?	M: I don't think I'd have been fined if I didn't ask someone for directions. I don't think I'd have been fined if I hadn't asked... (repeats, cut off)
LoRA r=4	Jack likes his poli-sci class. #Person1# thinks #Person2# will check it out.	#Person1# blames #Person2#'s fault for giving money to a policeman. #Person2# suggests driving instead of taking the train instead of taking the train.
LoRA r=16	Jack likes his poli-sci class. #Person1# thinks he will check it out.	#Person1# blames #Person2#'s fault for giving money to a policeman. #Person2# blames #Person1#'s fault because #Person1# didn't suggest driving instead of taking the train.

What changed. Before fine-tuning, the baseline copied a line from the dialogue or looped and repeated itself, so it never actually summarized. After LoRA fine-tuning, the model writes short third-person summaries in DialogSum's style with #Person1#/#Person2# tags. ROUGE-1 rose from 0.148 to 0.350 and ROUGE-L from 0.136 to 0.283. Content is still imperfect: Dialogue 1 says “Jack likes his poli-sci class” instead of the general topic of favorite classes, and Dialogue 2 muddles who blames whom.

4. LoRA rank experiment (r=4 vs. r=16)

r=16 has exactly 4x the trainable parameters of r=4 (688,128 vs. 172,032) and reached a slightly lower final training loss (1.450 vs. 1.534). On ROUGE the two are nearly identical (ROUGE-L 0.283 vs. 0.282; ROUGE-2 tied at 0.106). The visible difference was on Dialogue 2: r=4 produced a repetitive sentence (“instead of taking the train instead of taking the train”), while r=16 gave a coherent, complete second sentence. So the higher rank

gave a small gain in fluency and fit, but most of the improvement over the baseline came from LoRA fine-tuning itself, not from the extra rank. $r=4$ captures most of the benefit with a quarter of the parameters.

5. Limitations

ROUGE was computed on only 100 test dialogues with a single seed, so the $r=4$ vs. $r=16$ gap is within noise; the qualitative comparison rests on two samples. ROUGE also rewards word overlap rather than factual correctness, which is why fluent but slightly wrong summaries (as in Dialogue 2) still score well. A fairer rank comparison would use the full 499-dialogue test set and several seeds.

6. Reproducibility

Code: *DATA266_LoRA_DialogSum_final.ipynb* (Colab, T4). Adapters were saved to *outputs/lora_adapters_r4* and *outputs/lora_adapters_r16*. All numbers above appear in *METRICS.md* and *RUN_LOG.txt*. One environment issue occurred: an incompatible preinstalled torchao (0.10.0) blocked PEFT; upgrading torchao and restarting the session fixed it.